higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO SYSTEMS DEVELOPMENT

(Second Paper)

NQF LEVEL 2

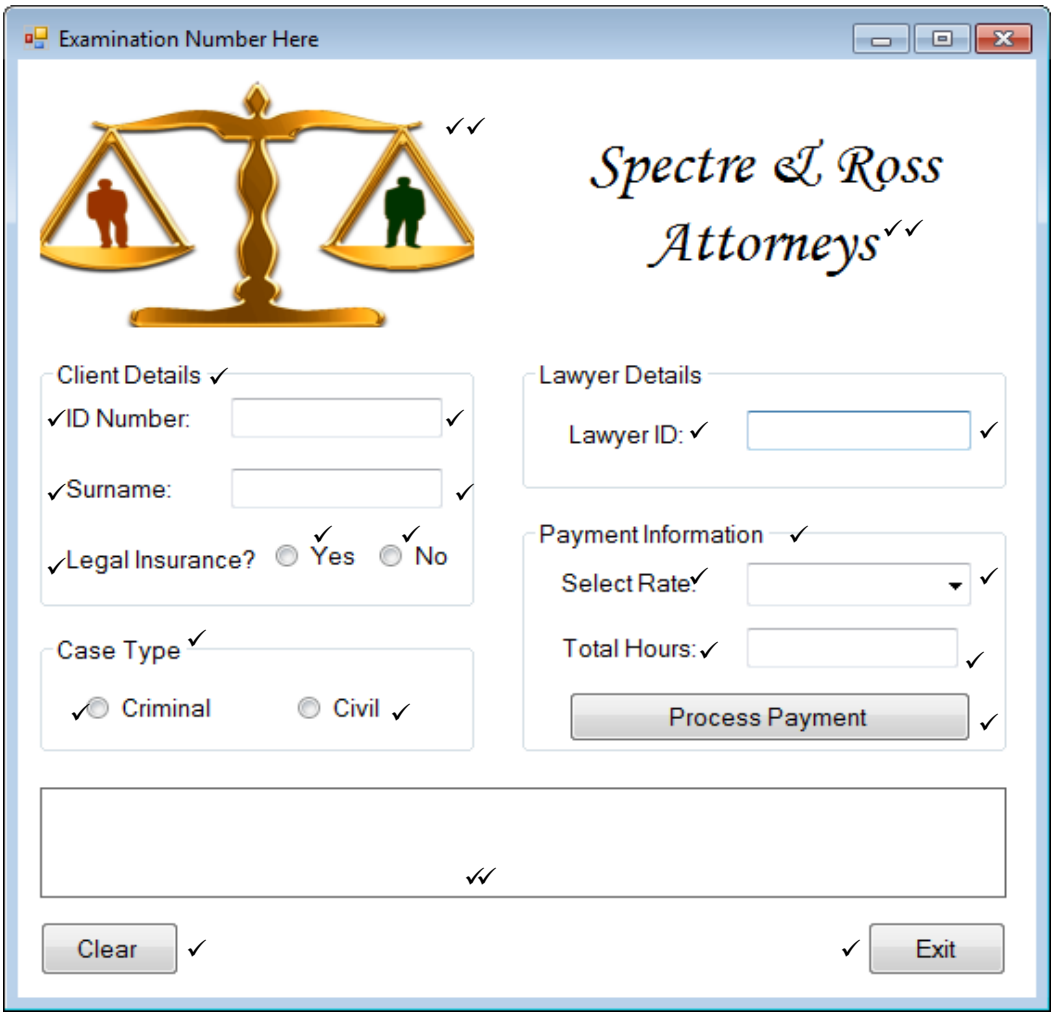
14 MARCH 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1-
1.8



- Examination Number + Printout of the form✓ **[28]**

QUESTION 2

2.1 2.1.1

```
cboConvertFrom.Items.Add("Rand")✓
cboConvertFrom.Items.Add("Bitcoin")✓

cboConvertTo.Items.Add("Rand")
cboConvertTo.Items.Add("Bitcoin")
```

(2 marks for either correct combo box)

2.1.2

```
cboConvertFrom.SelectedIndex = 0✓
cboConvertTo.SelectedIndex = 1✓
```

(2 × 2) (4)

2.2 2.2.1
```
Dim decAmountToConvert As Decimal = 0 ✓
Dim decConversionRate As Decimal = 0 ✓
Dim decConvertedValue As Decimal = 0 ✓
```
(3)

2.2.2
```
decAmountToConvert ✓ = CDec(txtValue.Text) ✓
```
(Or any other correct conversion method) (2)

2.2.3
```
If cboConvertFrom.SelectedIndex = 0 And
cboConvertTo.SelectedIndex = 1 Then ✓✓

    decConversionRate = 0.00002D ✓

ElseIf cboConvertFrom.SelectedIndex = 1 And
cboConvertTo.SelectedIndex = 0 Then ✓

    decConversionRate = 49820 ✓

Else
    decConversionRate = 1 ✓
End If
```
(NOTE: Students could have a variety of solutions that could work here. Markers should check code carefully.) (6)

2.2.4
```
decConvertedValue = decAmountToConvert✓ *✓ decConversionRate✓
```
(3)

2.2.5
```
lblDisplay.Text = decAmountToConvert.ToString("F2") & " " & _ ✓
cboConvertFrom.SelectedItem.ToString & " = " & _
decConvertedValue.ToString("F2") & " " & _ ✓
cboConvertTo.SelectedItem.ToString ✓
```
(3)

2.3
```
txtValue.Clear()✓
lblDisplay.Text = "" ✓
cboConvertFrom.SelectedIndex = 0 ✓
cboConvertTo.SelectedIndex = 1 ✓
```
(4)

2.4
```
Me.Close() ✓
```
(1)
[26]

QUESTION 3

3.1 3.1.1
```
Dim decTileCost As Decimal
Dim decLength As Decimal
Dim decBreadth As Decimal
Dim decArea As Decimal
Dim decTotalCost As Decimal
```
(Any THREE correct variables) (3)

3.1.2
```
decLength = CDec(txtLength.Text) ✓
decBreadth = CDec(txtBreadth.Text) ✓
```
(2)

3.1.3
```
decArea = decLength✓ *✓ decBreadth✓
```
(3)

3.1.4
```
If radStandard.Checked = True Then ✓
    Dim intAnswer As Integer
    intAnswer = MessageBox.Show("Are you sure you don't want the
    extra strength tiles?", "Better Option", MessageBoxButtons.YesNo,
    MessageBoxIcon.Question) ✓
    If intAnswer = vbNo Then ✓
         Exit Sub ✓
    End If
End If
```
(Or any other correct method used by student) (4)

3.1.5
```
If radStandard.Checked = True Then ✓
        decTileCost = 15 ✓
ElseIf radExtra.Checked = True Then
        decTileCost = 20 ✓
End If
```
(3)

3.1.6
```
decTotalCost = decArea✓ *✓ decTileCost✓
```
(3)

3.1.7
```
lblDisplay.Text = "The job will cost: "✓ & _
decTotalCost.ToString("C2") ✓
```
(2)

3.2
```
radStandard.Checked = False  }
radExtra.Checked = False     } ✓
txtLength.Clear()  }
txtBreadth.Clear() } ✓
lblDisplay.Text = "" ✓
```
(3)

3.3
```
Me.Close() ✓
```
(1)
[24]

QUESTION 4

4.1-4.2
```
Public Class Form1
      'Examination Number ✓
    Private Sub btnCompare_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnCompare.Click
        'Player 1 variables
        Dim strP1Surname As String = "" ✓
        Dim intP1Games As Integer = 0
        Dim intP1Runs As Integer = 0
        Dim intP1Wickets As Integer = 0
        Dim decP1BattingAverage As Decimal = 0
        Dim decP1BowlingAverage As Decimal = 0

        'Player 2 variables
        Dim strP2Surname As String = ""
        Dim intP2Games As Integer = 0 ✓✓
        Dim intP2Runs As Integer = 0
        Dim intP2Wickets As Integer = 0
        Dim decP2BattingAverage As Decimal = 0
        Dim decP2BowlingAverage As Decimal = 0
```

```
        'Averages ✓  (make comment)
        Dim strBetterBowler As String = ""
        Dim strBetterBatsman As String = ""

        strP1Surname = txtP1Surname.Text
        intP1Games = CInt(txtP1NoGames.Text)
        intP1Runs = CInt(txtP1Runs.Text)
        intP1Wickets = CInt(txtP1Wickets.Text)

        strP2Surname = txtP2Surname.Text
        intP2Games = CInt(txtP2NoGames.Text)
        intP2Runs = CInt(txtP2Runs.Text)
        intP2Wickets = CInt(txtP2Wickets.Text) ✓

        'Calculate Averages
        decP1BattingAverage = CDec(intP1Runs / ✓✓intP1Games)
        decP1BowlingAverage = CDec(intP1Wickets / intP1Games)

        decP2BattingAverage = CDec(intP2Runs / intP2Games)
        decP2BowlingAverage = CDec(intP2Wickets / intP2Games)

        'Find better batsmen
        If decP1BattingAverage >✓✓ decP2BattingAverage Then (change to >)
            strBetterBatsman = strP1Surname
        ElseIf decP1BattingAverage < decP2BattingAverage Then
            strBetterBatsman = strP2Surname
        Else
            strBetterBatsman = "They are equal"
        End If

        'Find better bowler
        If decP1BowlingAverage > decP2BowlingAverage Then
            strBetterBowler = strP1Surname
        ElseIf decP1BowlingAverage < decP2BowlingAverage Then
            strBetterBowler = strP2Surname
        Else
            strBetterBowler = "They are equal"
        End If

        'Display Comparisons
        lstDisplay.Items.Add("Stat".PadRight(15) & "|  " & "Player
1".PadRight(15) & "|  " & "Player 2")
        lstDisplay.Items.Add("-----------------------------------------------
---------")
        lstDisplay.Items.Add("Surname".PadRight(15) ✓✓ & "|  " &
strP1Surname.PadRight(15) & "|  " & strP2Surname)
        lstDisplay.Items.Add("Games:".PadRight(15) & "|  " &
intP1Games.ToString.PadRight(15) & "|  " & intP2Games.ToString)
        lstDisplay.Items.Add("Runs:".PadRight(15) & "|  " &
intP1Runs.ToString.PadRight(15) & "|  " & intP2Runs.ToString)
        lstDisplay.Items.Add("Wickets:".PadRight(15) & "|  " &
intP1Wickets.ToString.PadRight(15) & "|  " & intP2Wickets.ToString)
        lstDisplay.Items.Add("Bat Average:".PadRight(15) & "|  " &
decP1BattingAverage.ToString("F2").PadRight(15) & "|  " &
decP2BattingAverage.ToString("F2"))
        lstDisplay.Items.Add("Bowl Average:".PadRight(15) & "|  " &
decP1BowlingAverage.ToString("F2"✓✓).PadRight(15) & "|  " & (change to "F2")
decP2BowlingAverage.ToString("F2"))
```

```
        'Display better batsman and bowler
        lblDisplay.Text = "The better batsman is: " & strBetterBatsman &
ControlChars.NewLine & _
                          "The better bowler is: " & strBetterBowler

    End Sub ✓


    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnClear.Click
        'Clear textboxes
        txtP1NoGames.Clear()
        txtP1Runs.Clear()
        txtP1Surname.Clear()
        txtP1Wickets.Clear()

        txtP2NoGames.Clear()
        txtP2Runs.Clear()
        txtP2Surname.Clear()
        txtP2Wickets.Clear()

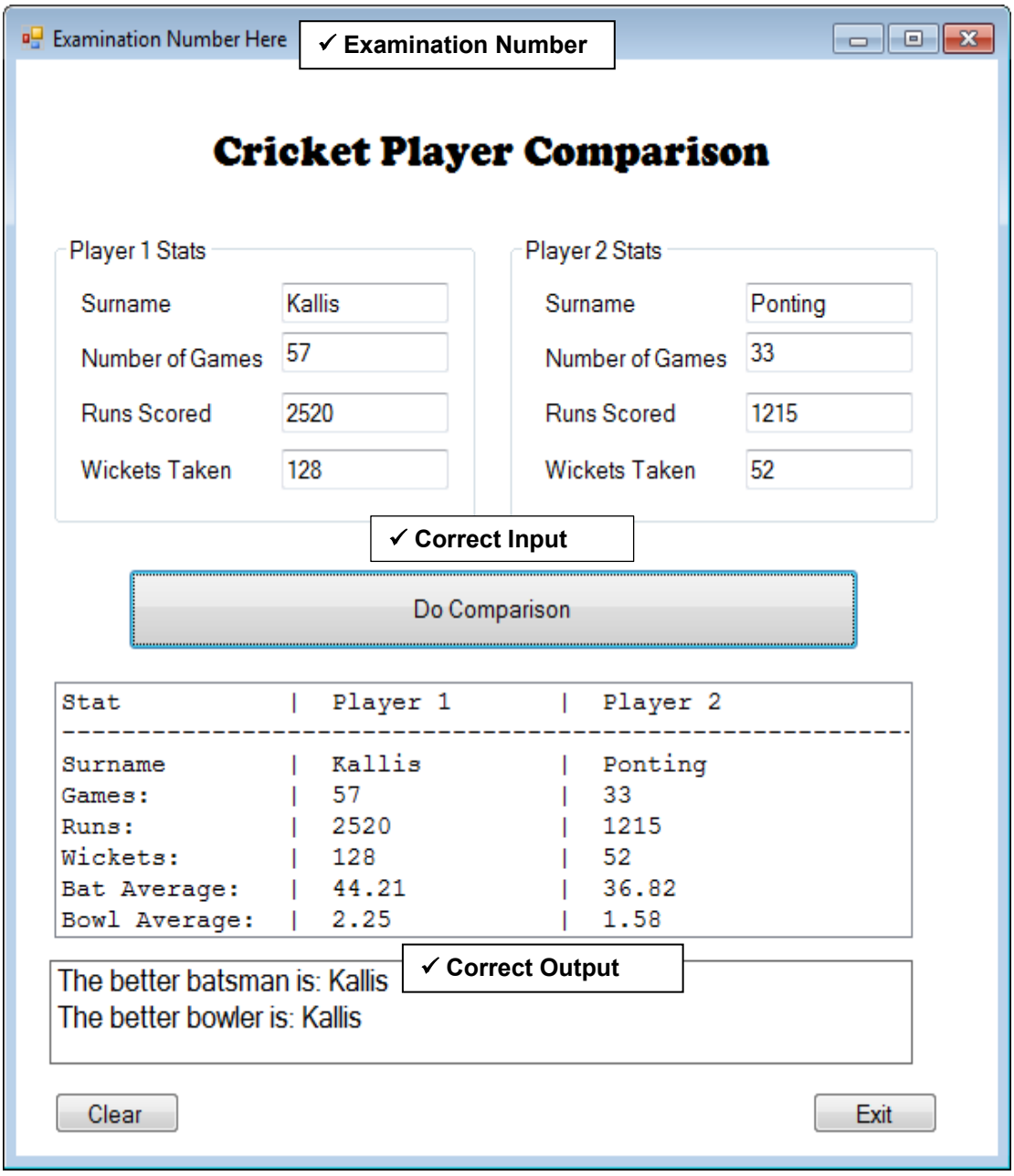
        'Clear label and listbox
        lstDisplay.Items.Clear()✓
        lblDisplay.Text = ""


    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnExit.Click
        'Exit program
        Dim intAnswer As Integer
        intAnswer = MessageBox.Show("Are you sure you want to exit?", "Exit",
MessageBoxButtons.OKCancel, MessageBoxIcon.Question)
        If intAnswer = vbOk ✓✓ Then
            Me.Close()
        End If
                          ✓ Printout of Source Code
    End Sub
```

(19)

4.3



(3)
[22]

TOTAL: 100

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